

Payton Suh

Department of Mathematics, UCLA · paytonsuh@ucla.edu · paytonsuh.com

EDUCATION

University of California, Los Angeles

2026–

B.S. Mathematics, expected June 2028

M.A. Mathematics, expected June 2029 (Departmental Scholars Program, intended)

Santa Monica College

2024–2026

A.S.-T. Mathematics; A.S. General Science. GPA 3.85/4.00; Dean's List 2024–2026. Began college at sixteen.

COURSEWORK

UCLA, in progress

Algebra (110AH), Linear Algebra (115AH), Analysis (131AH), all honors

UCLA, planned

Abstract Algebra, Analysis, Complex Analysis, Geometric Analysis, Topology, Probability Theory, Stochastic Calculus

Santa Monica College

Calculus I–III, Linear Algebra, Differential Equations; Engineering Physics I–III, Modern Physics, Chemistry, Circuit Analysis; Programming (Python, C++, Data Structures, MATLAB), Data Science

RESEARCH

ZrC-ARC: Zirconium Carbide Architected Refractory Coatings

2026–

Grain morphology and hydrogen resistance of ZrC coatings for hypersonic and nuclear thermal propulsion. Advisor: Prof. Daniel R. Mumm, UC Irvine Materials Research Institute. \$10,000 seed grant, NASA L'SPACE Program (Spring 2026).

Co-Investigator

Oct 2026–

Leads research and LAMMPS molecular-dynamics simulation.

Principal Investigator

Jan–Oct 2026

Founded the project, built the team, and wrote the winning proposal.

Stellar Light Curve Analysis

2025–2026

Python and machine-learning pipeline for public survey light curves. Co-author of a research submission reporting a previously unreported flaring variable star.

TEACHING

Student Instructor, SMC Science Tutoring Center

2026–

Selected by the STEM department to tutor physics and chemistry; 10+ sessions weekly.

Mentor and Lead Builder, VEX Robotics (11444X, 334R)

2023–

Mentored teams to VEX Worlds Division Semifinals and the California Region 4 State Tournament; design, controls, and autonomous programming.

Taekwondo Instructor, Elite Taekwondo Center

until 2024

Kukkiwon-certified third-degree black belt.

SERVICE

Vice President of California, Back to School Together

2025–

Raised over \$1,200 for STEM education for students in Mexico and Guatemala.

SKILLS

Computing

Python, C++, MATLAB, Simulink, SQL, LaTeX

Engineering

SolidWorks, CAD, circuit analysis, data analysis and visualization

Languages

English (native), Korean (native), Japanese (advanced)

OTHER

First violin, YOLA HOLA Symphony Orchestra and Torres Chamber Ensemble (2018–2024). Interests: bouldering (V7), billiards (8-ball and 9-ball), cooking, photography.